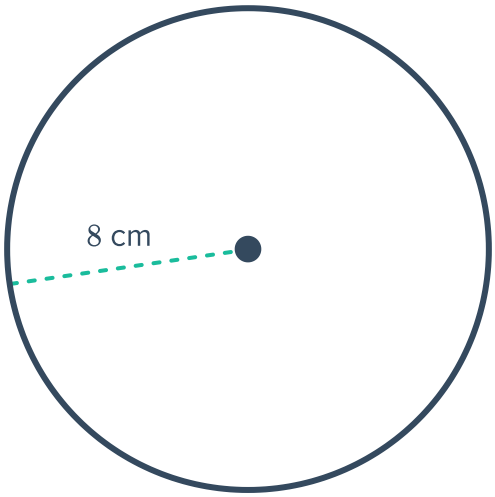
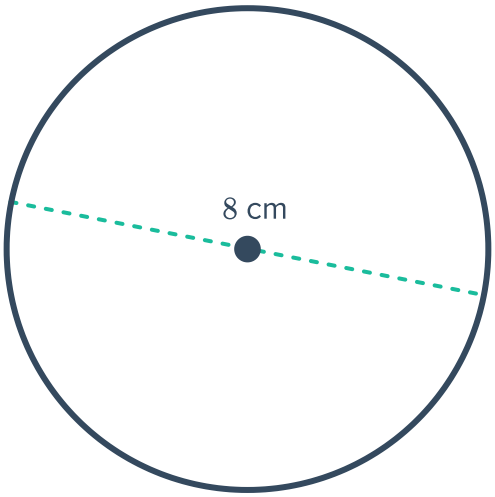
Circumference of a circle

1 Find the circumference of the following circles to two decimal places:

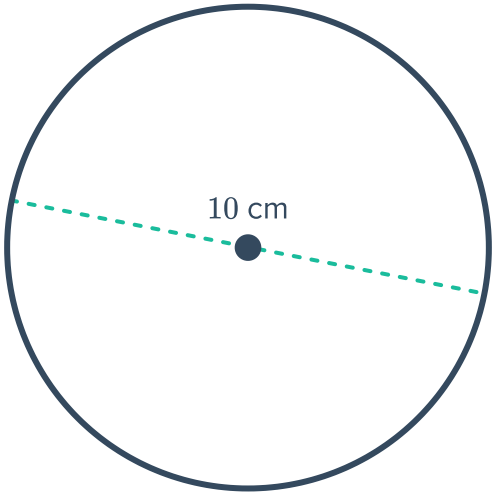
a



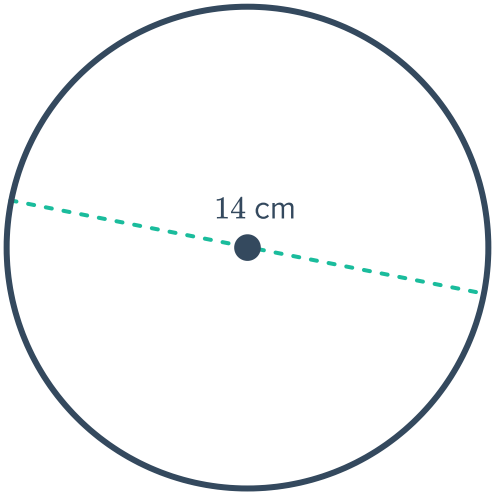
b



c



d



2 If the radius of a circle is 27 cm, find its circumference. Round your answer to one decimal place.

3 If the diameter of a circle is equal to 38 cm, find its circumference. Round your answer to one decimal place.

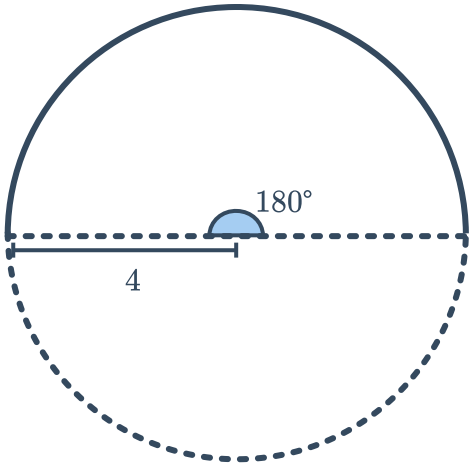
4 Find the radius of a circle with a circumference of 14 cm. Round your answer to two decimal places.

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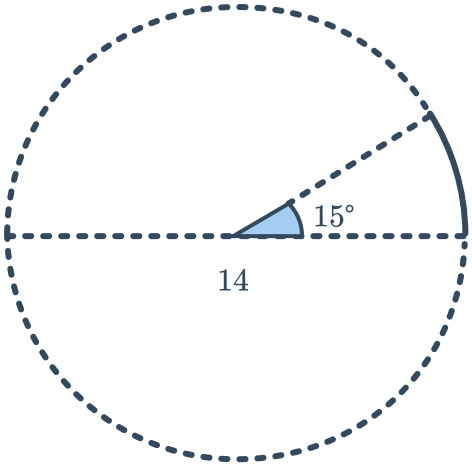
Arc length

5 For each of the following sectors, find the length of the arc to one decimal place:

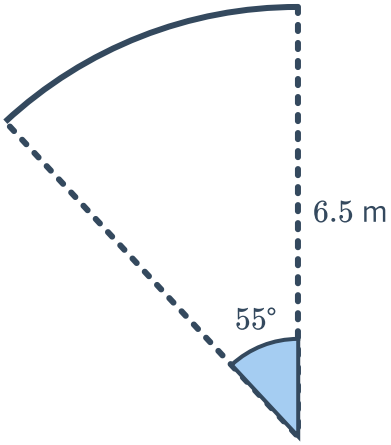
a



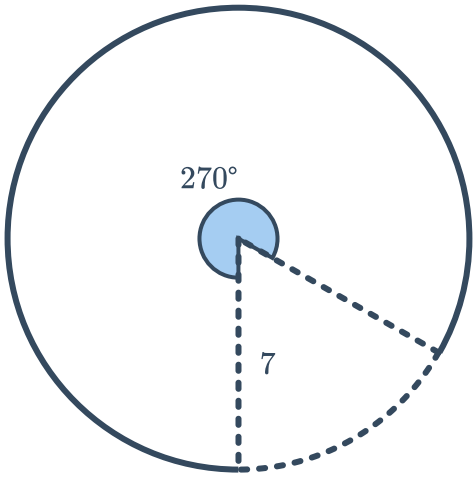
b



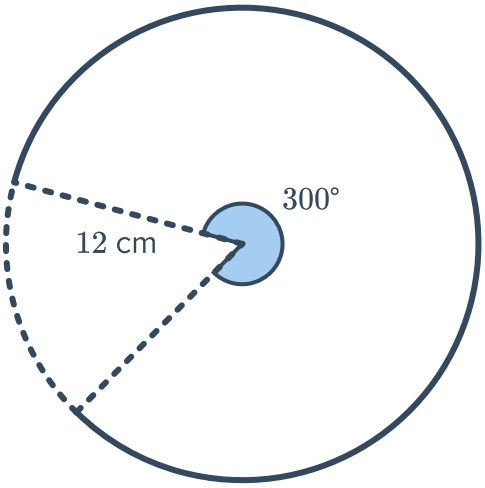
c



d



e



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6 The length of an arc that subtends an angle of 29° at the centre of a circle is 10 cm. Find the radius of the circle correct to two decimal places.

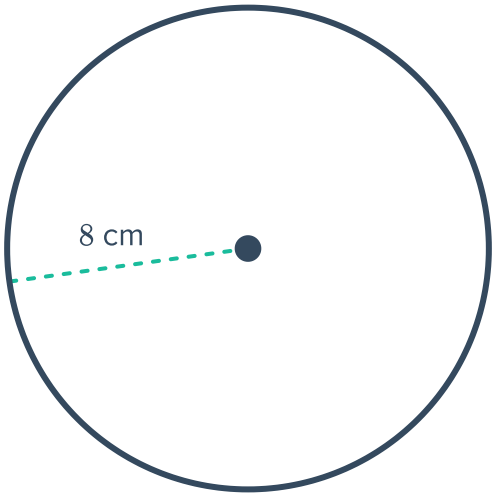
7 The arc formed by two points on a sphere with a radius of 2 m subtends an angle of 37° at the centre. Find the length of the arc correct to two decimal places.

8 An arc formed by two points on Saturn's surface subtends an angle of 25° at Saturn's centre. The radius of Saturn is $58\,232$ km. What is the length of the arc correct to 0.1 of a kilometre?

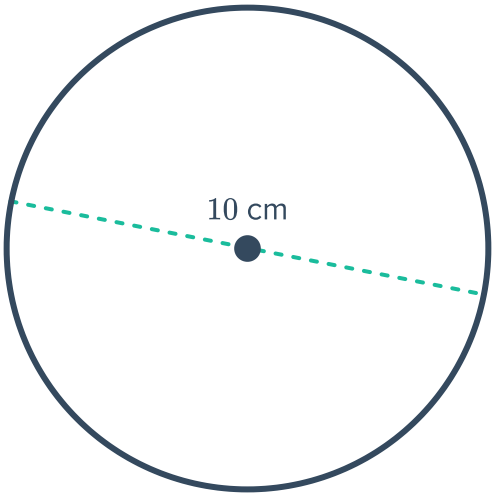
Area of circle

9 Find the area of the following circles to one decimal place:

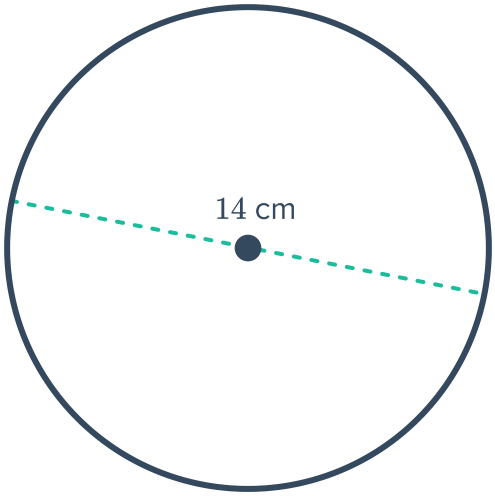
a



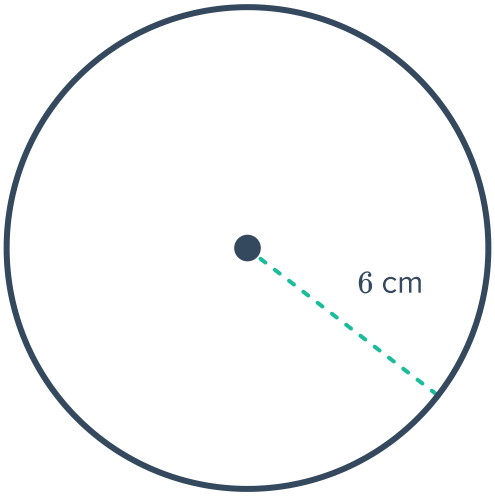
b



c



d



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10 The area of a circle is $352\ \text{cm}^2$.

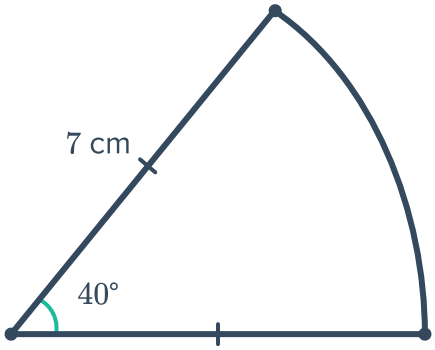
a If its radius is r cm, find r, round your answer to two decimal places.

b Using the rounded value from the previous part, find the circumference of the circle. Round your answer to one decimal place.

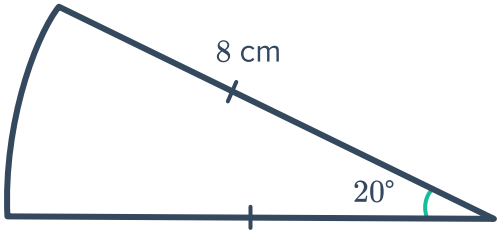
Sectors

11 Find the perimeter of the following sectors. Round your answers to one decimal place.

a



b

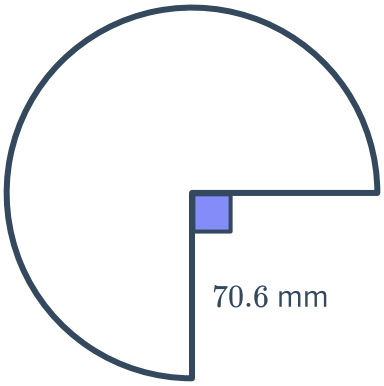


12 For each sector below, calculate the following to two decimal places:

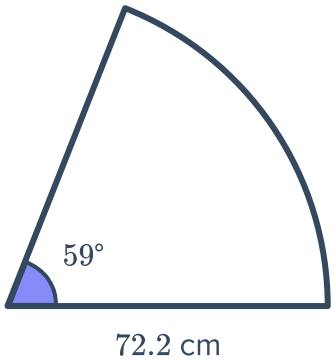
i The perimeter

ii The area

a

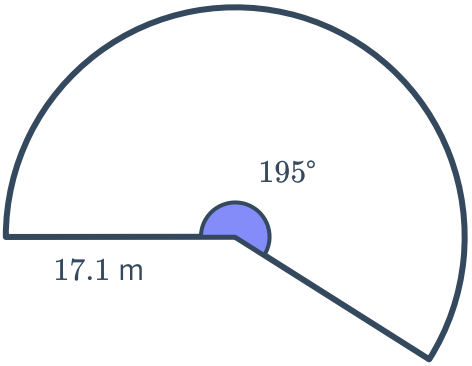


b

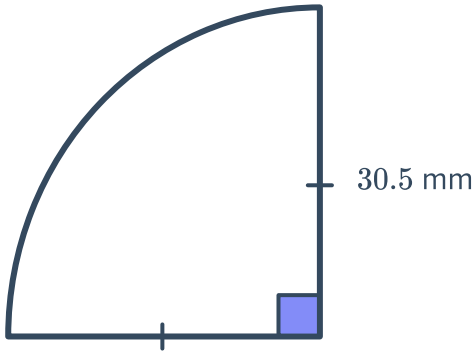


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c



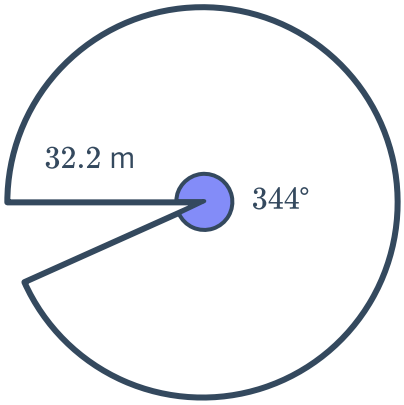
d



13 Consider the following sector. The marked angle is 344°.

a Calculate the perimeter of the sector. Round your answer to two decimal places.

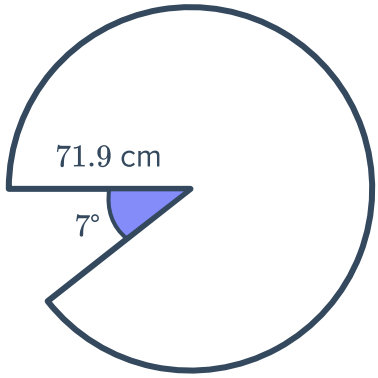
b Calculate the area of the sector. Round your answer to two decimal places.



14 Consider the following sector::

a Calculate the perimeter. Round your answer correct to two decimal places.

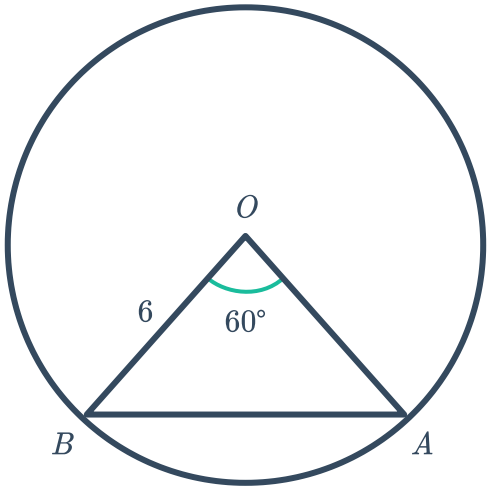
b Calculate the area. Round your answer correct to two decimal places.



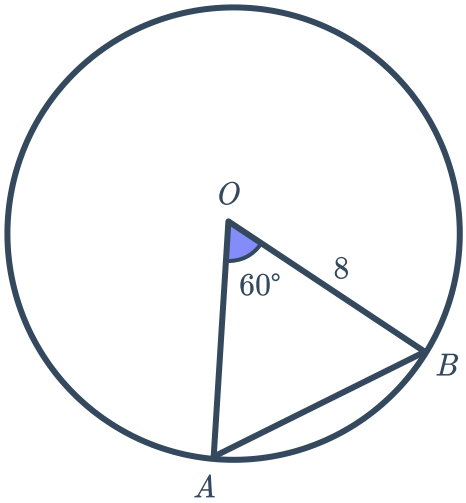
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Segment

15 The diagram shows a circle with a radius of 6 units, and chord AB subtending an angle of 60° at the centre. Find the exact area of the minor segment cut off by chord AB.



16 The diagram shows a circle with radius 8 units, and chord AB subtending an angle of 60° at the centre.
Find the exact area of the major segment cut off by chord AB.



17 Consider a circle with centre O and a chord AB subtended by an angle of θ degrees at the centre. The radius is 36 cm and the area of sector OAB is 108π cm^2.

- **a** Find θ.
- **b** Find the exact area of the minor segment cut off by chord AB.
- **c** Find the exact area of the major segment cut off by chord AB.
- **d** Find the exact ratio of the area of the major segment to the area of the minor segment.

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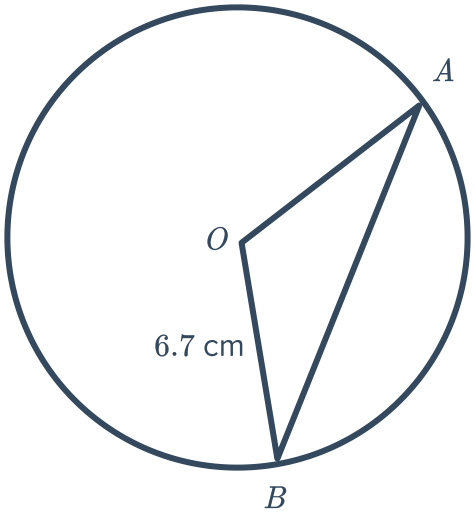
18 A chord AB of a circle with centre O has a length of 6 cm. The radius of the circle is 5 cm.

a Find the size of acute $\angle AOB$. Round your answer to one decimal place.

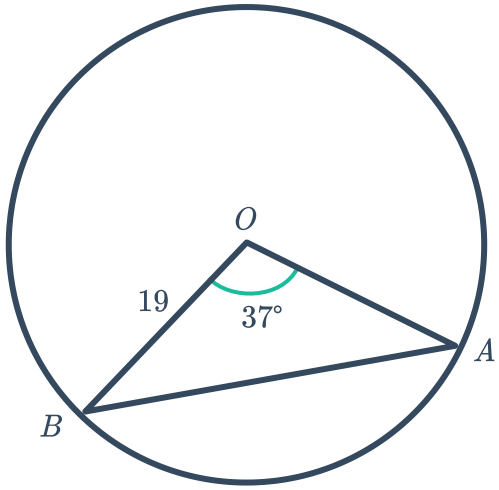
b Find the length of the minor arc AB.

c Find the area of the minor segment formed by the chord AB. Round your answer to one decimal place.

19 In the diagram, O is the centre of the circle, and sector OAB takes up $\frac{3}{7}$ of the circle. Find the area of the minor segment cut off by chord AB. Round your answer to one decimal place.



20 The diagram shows a circle with radius 19 mm, and chord AB subtending an angle of 37° at the centre. Find the exact area of the minor segment cut off by chord AB.



21 A circle with centre O has an arc AB of length 12 cm subtended by an angle of θ at the centre. The radius of the circle measures 9 cm.

a Find θ, the angle subtending the arc AB.

b Find the area of the minor segment cut off by chord AB. Round your answer to one decimal place.

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Applications

22 A circular metal plate is cut into two segments along a chord equal in length to the radius. Let r be the radius of the circle, and θ be the angle at the centre subtending the chord of length r.

a Find θ.

b What is the ratio of the area of the larger segment to the area of the smaller one?

23 Two identical circles of radius length 7 cm intersect at points A and B, and have their centres 12 cm apart.

a C is the midpoint of chord AB, and is x cm away from each center. Find x.

b Find θ, the angle at the centre of each circle that subtends chord AB. Round your answer to two decimal places.

c Calculate the area common to each circle. Round your answer to one decimal place.

24 Consider a circle with centre O and a chord AB subtended by an angle of 60° at the centre. The area of the minor segment cut off by chord AB is $24\pi - 36\sqrt{3}\ \text{cm}^2$.

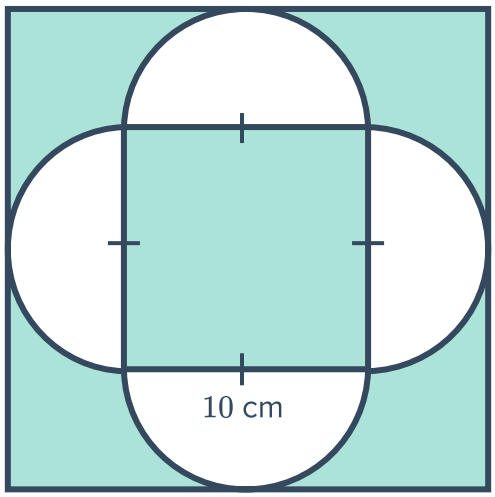
a Find r, the radius of the circle.

b Find the arc length AB. Leave your answer in terms of π.

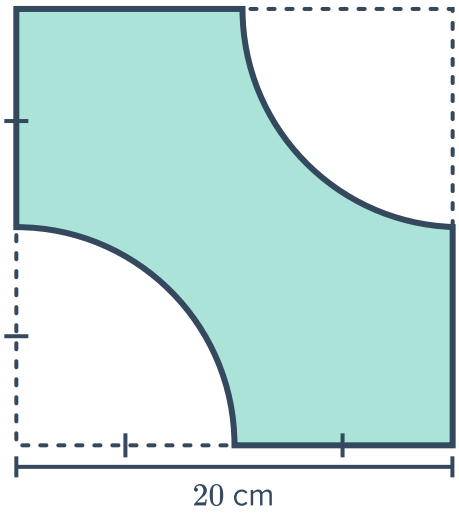
25 A circular running track has a diameter of 23 metres. How many laps must be completed to run 1600 metres? Round your answer to one decimal place.

26 Find the area of the shaded region in the following figures. Round your answers to one decimal place.

a

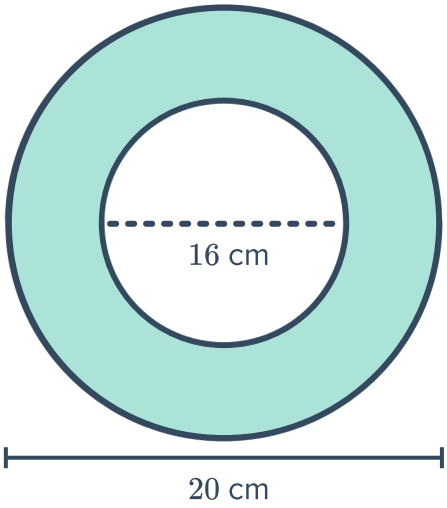


b

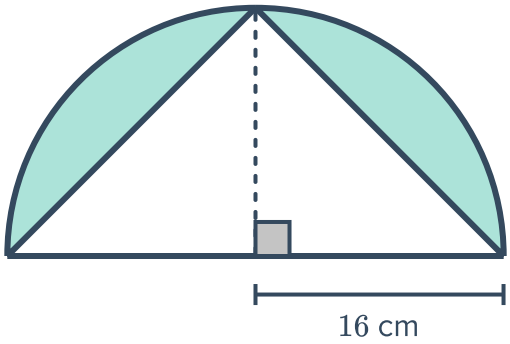


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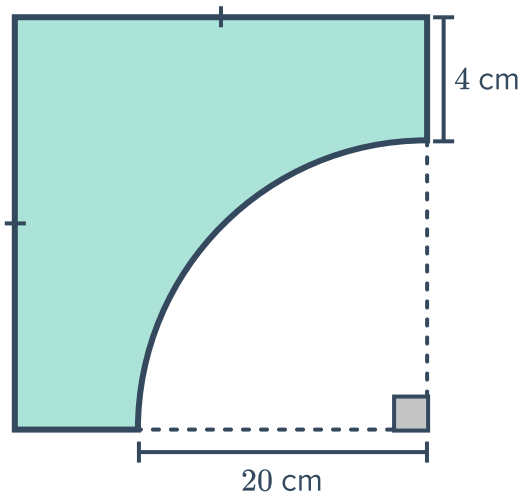
c



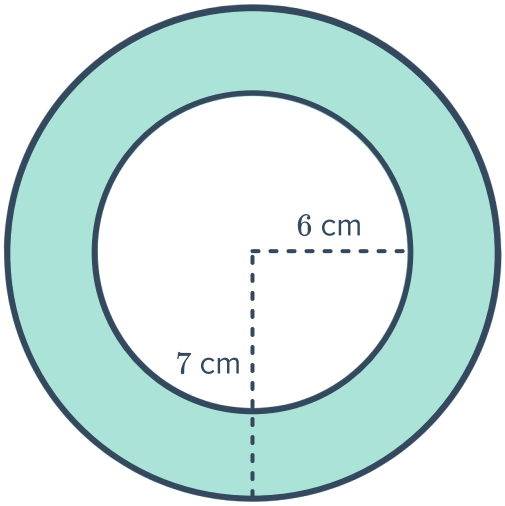
d



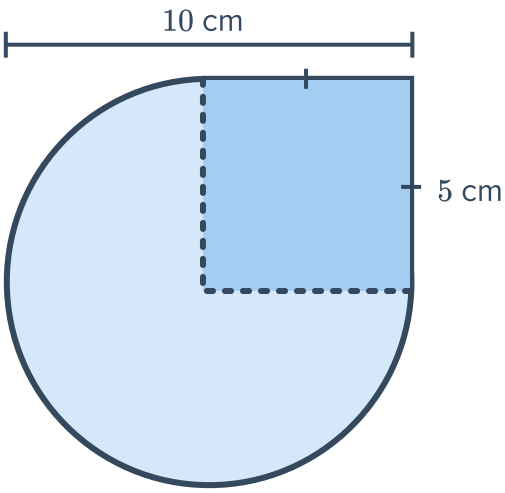
e



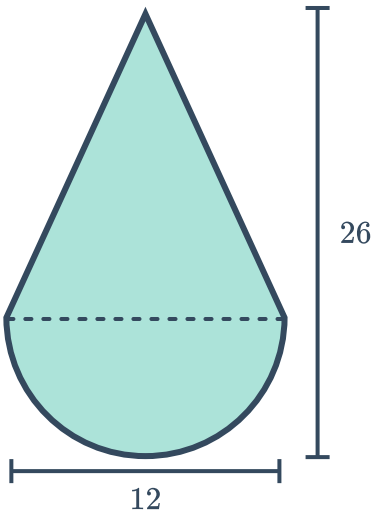
f



g

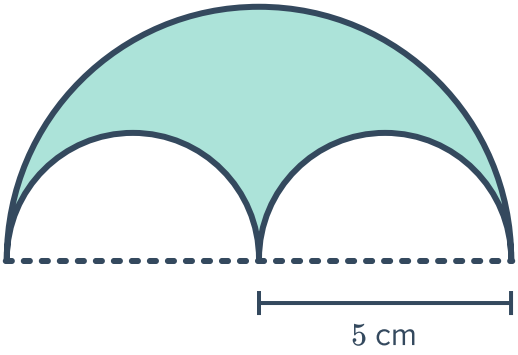


h

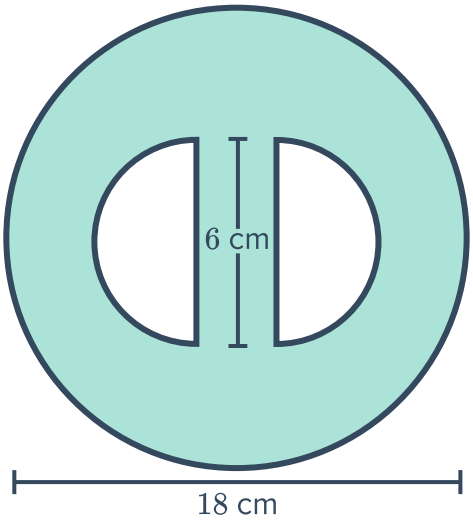


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i

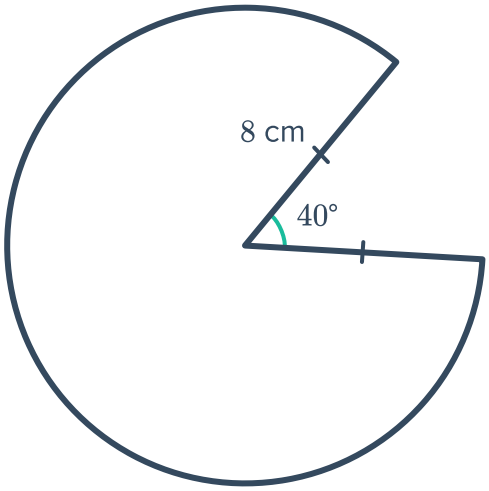


j

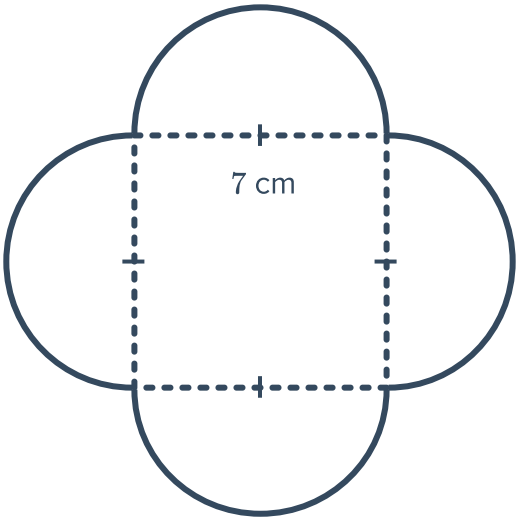


27 Find the perimeter of the figures shown below. Round your answers to two decimal places.

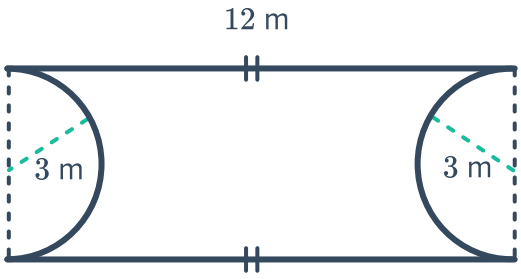
a



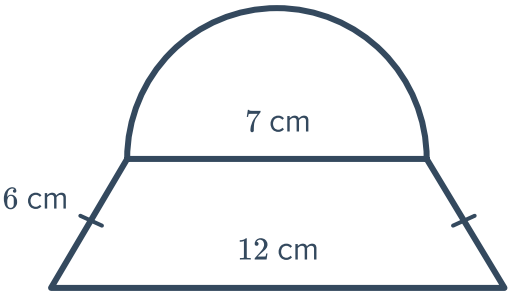
b



c

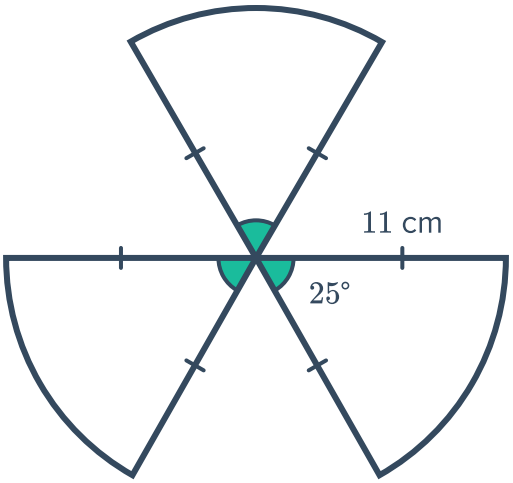


d

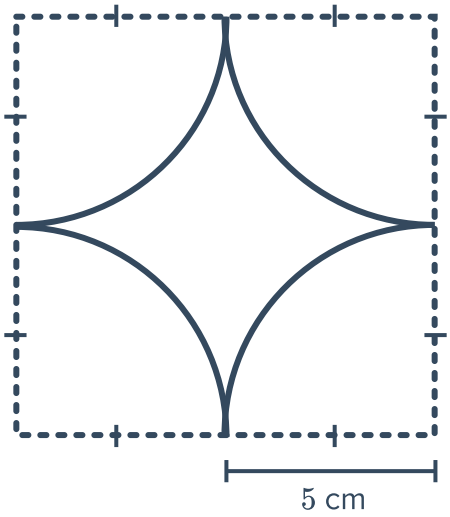


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e

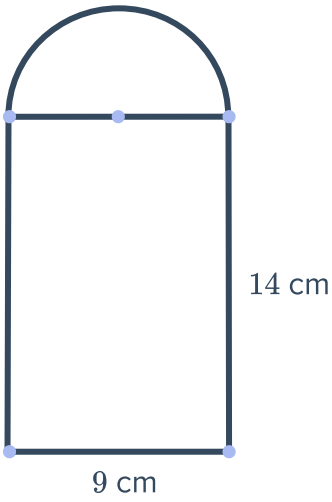


f

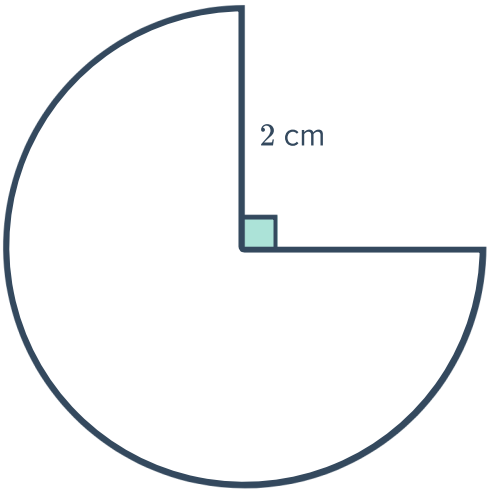


28 Find the area of the following figures. Round your answers to one decimal place.

a



b

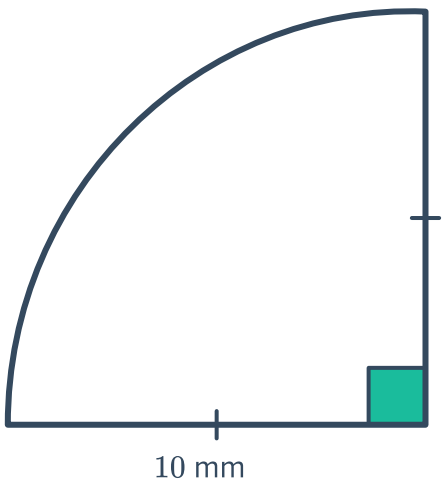


29 Calculate the perimeter of the following figures. Round your answers to one decimal place.

a



b



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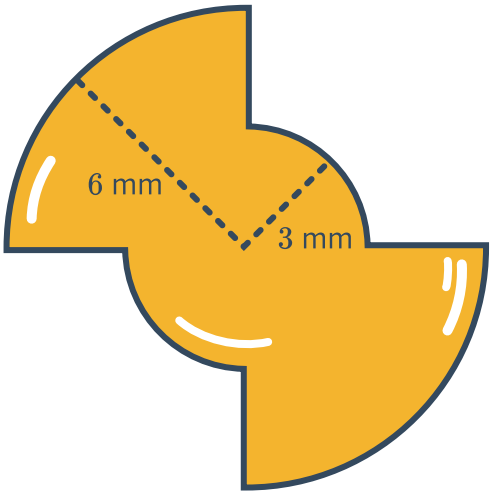
30 A dining table is in the shape of a rectangle plus two semicircles, as shown in the figure. What is the perimeter of the table in meters, corrected to the nearest 0.1 metre?



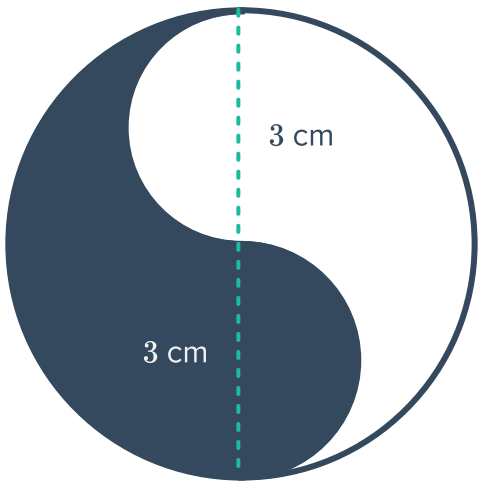
31 The diagram shows a piece of jewellery made out of gold:

a Find the area of the piece. Round your answer to the nearest whole number.

b If the gold costs $\$4$ per square millimetre, find the cost of the piece of jewellery.

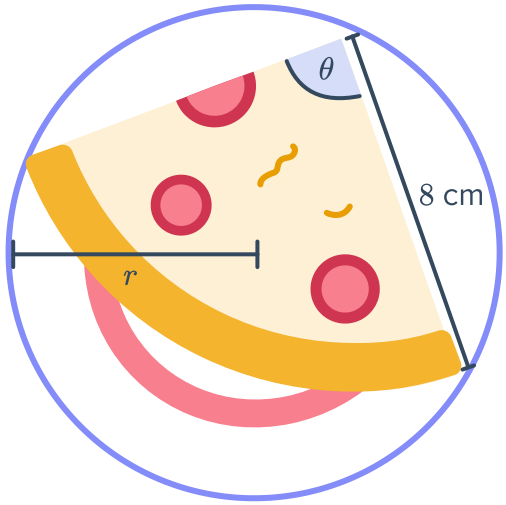


32 The design attached is made using a large circle and two smaller circles of diameter 3 cm. Find the area of the shaded region correct to one decimal place.



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33 A circular pizza of radius 8 cm is cut into sectors. Each sector is to be placed on a circular plate of radius r cm, that is just large enough to contain that sector. A sector of pizza is cut where the angle at the center is θ, with $0 < \theta < 90^\circ$. It is placed on a circular plate as shown below:



a If the slice of pizza is $\frac{1}{5}$ of the whole pizza, find the area of the slice of pizza. Round your answer to one decimal place.

b Find the radius, r, of the plate that is needed to hold the pizza. Round your answer to one decimal place.

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